

AI+RES: forecasting rare weather extremes three weeks out, at a fraction of the compute

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Abstract

Three weeks before an extreme weather event, a conventional forecast ensemble of 24 members does not produce it. The event is rarer than 1 in 24, so the ensemble returns no member at the observed value and the forecast carries no information about the tail. We report results from AI+RES, a rare-event sampling method that runs 64 forecast trajectories and resamples them five times over the forecast, discarding the trajectories that do not approach the event and duplicating those that do. The resampling is driven by a second, independently trained AI weather model, which forecasts each trajectory’s outcome at the target week and ranks the population by it. On the seven test cases at or beyond the 96th percentile of the 1959 to 2022 climatological distribution, AI+RES produced forecast members that reached the observed anomaly in six cases. Direct ensemble sampling reached it in none of the seven, and the operational Climate Forecast System version 2 (CFSv2) in none of the seven. One AI+RES forecast takes 7.2 to 8.7 hours and 61 to 64 H100 GPU-hours on a single eight-GPU node, about \$155 at \$2.50 per GPU-hour. Obtaining the same set of extreme trajectories by direct sampling is estimated at 5 to 23 times that cost, and the multiple grows with the rarity of the event. Nine test cases support a qualitative assessment. They are not enough to establish a hit rate, a skill score, or a calibrated probability, and no such claim is made here. Every number below is measured from a run unless it is marked as an estimate.

1 The problem

Weather forecasts are run as ensembles. The same model is launched many times from slightly different starting points, and the spread of the outcomes is the forecast. The number of runs sets the smallest probability the ensemble can resolve. With 24 runs, nothing rarer than about 1 in 24 can be distinguished from zero, and a single run landing on the event is indistinguishable from chance.

Three weeks out, the events that matter most are the ones in that blind spot. The 2021 Pacific Northwest heat dome ran 7.72 K above normal over its region for a week, which for that region and season is about a 1-in-22-year week (Table 1). A 24-member GenCast ensemble launched three weeks earlier produced no run that reached it. CFSv2 at the same lead forecast a week near normal. Neither model is wrong about the atmosphere. Both are being asked a question their sample size cannot answer.

2 The method

Rare-event sampling concentrates a fixed compute budget on the trajectories that approach the event, instead of spreading it evenly over the forecast distribution. We launch 64 forecast trajectories from the same starting conditions. Every three days we stop them, score each one, discard the low scorers, and duplicate the high scorers in their place. The population stays at 64 members throughout and is progressively concentrated on the trajectories heading toward the extreme. Weights carried alongside each trajectory convert the concentrated population back into an unbiased probability at the end.

The score is the step that decides the outcome. The conventional choice is the trajectory’s current temperature, which three weeks before a heat wave carries no information about which trajectory becomes hot. We instead score each trajectory by running a second, independently trained AI weather model forward from it and taking its forecast for the target week. At the 9-day checkpoint of the 2021 heat dome, the AI score ranks the trajectories

with a rank correlation of 0.80 against where they in fact ended up. The conventional score reaches 0.06 at the same checkpoint. Run on the identical setup with only the score exchanged, the conventional method reaches the observed 2021 heat dome with 0 of 64 trajectories.

The trajectory model is GenCast (Google DeepMind) at 0.25-degree resolution. The scoring model is FourCastNet 3 (NVIDIA). Both were trained on data ending before 2019, so every event tested here is out of sample.

3 Results

Table 1: Nine test cases, each forecast from three weeks before the event. A_L is the event’s temperature anomaly, the week-long average over the event’s region measured against the 1990 to 2019 normal, in kelvin (1 K = 1.8 °F). Each method column gives that method’s most extreme member, which is the closest any of its runs came to the event. **Bold** marks a method that reached or passed the observed value. Percentile is the event’s rank against the 1959 to 2022 reanalysis record for the same region and calendar season, with the return period estimated from a generalised extreme value fit in parentheses. Lead times differ by column and are given in the text below.

		A_L , week-mean temperature anomaly (K)			
event	percentile	observed	AI+RES	CFSv2	direct sampling
<i>Extremes</i>					
S. Texas heat dome, Jun 2023	>99.96 (no analogue)	+5.73	+6.19	+1.19	+4.19
Southwest heat wave, Aug 2020	99.96 (1-in-210 y)	+5.14	+4.31	+1.60	+3.82
California heat wave, Sep 2022	99.71 (1-in-37 y)	+4.71	+6.33	+3.68	+4.70
Pacific NW heat dome, Jun 2021	99.60 (1-in-22 y)	+7.72	+10.29	+3.42	+7.08
CONUS warm week, Dec 2025	98.78 (1-in-10 y)	+3.26	+5.12	+0.22	+3.46
Winter Storm Elliott, Dec 2022	98.41 (1-in-5 y)	−14.42	−17.04	−10.74	−9.83
Winter Storm Uri, Feb 2021	96.80 (1-in-4 y)	−7.40	−10.20	−3.86	−6.51
<i>Mild controls</i>					
CONUS warm week, Aug 2024	94.08 (1-in-4 y)	+1.06	+2.37	+1.35	+1.96
CONUS median week, Nov 2023	60.10 (typical)	+0.21	+4.16	+1.75	+3.14
reached, 7 extremes			6 of 7	0 of 7	0 of 7
reached, all 9			8 of 9	2 of 9	3 of 9

Lead times are not identical across the columns of Table 1. AI+RES forecasts from a 21-day lead. CFSv2 is also at 21 days, as four operational cycles launched 6 hours apart. Direct sampling is a 24-member GenCast ensemble at a 14-day lead in every row except the S. Texas case, where it is 16 GenCast runs at 21 days. A separate 24-member FourCastNet 3 ensemble at 21 days was run on five of the nine cases and reached the observed value on one of them, the 2021 heat dome, with one member. The direct-sampling column therefore had a one-week shorter lead than AI+RES in most rows, which favours it, and it still reached none of the seven extremes.

Rarity is what separates the methods. Where the event is rare, AI+RES produces members that reach it and the two alternatives do not. Where the event is ordinary, which is the case for the two controls in Table 1, all three methods reach it. The controls are the check that the difference between methods is a function of rarity and not of one model being more accurate in general. The single miss, the 2020 Southwest heat wave, is the rarest case in the set by return period. Resampling moved the population 1.75 K toward the observed value and stopped 4.4 K short of it.

The bound on what this configuration reaches is qualitative. On the evidence of these nine cases, the configuration tested here, 64 trajectories with five resampling steps at a 21-day lead, produces the event for extremes up to about the 99.9th percentile, which is roughly a 1-in-40-year week for a given region and season. It reached one event with no analogue in the 64-year record, the S. Texas heat dome, and missed another, the Southwest heat wave at an estimated 1-in-210-year return period. The boundary for this budget is therefore near the 1-in-100-year

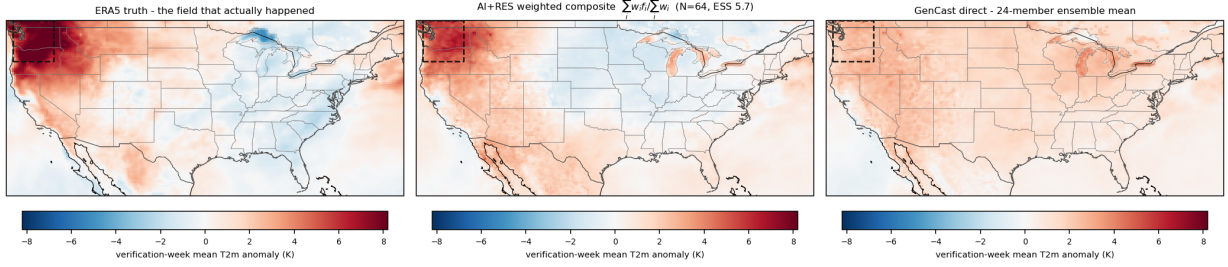


Figure 1: The 2021 Pacific Northwest heat dome, as maps of the week-average temperature anomaly. **Left:** the observed field. **Middle:** the AI+RES population three weeks out, with the dome in the observed location at the observed intensity. **Right:** the 24-member direct GenCast ensemble average from the same starting point, which contains no dome. The dashed box is the scored region. The middle panel is a weighted composite of trajectories selected for this event, so it shows where the method placed its samples and is not a forecast in the ordinary sense.

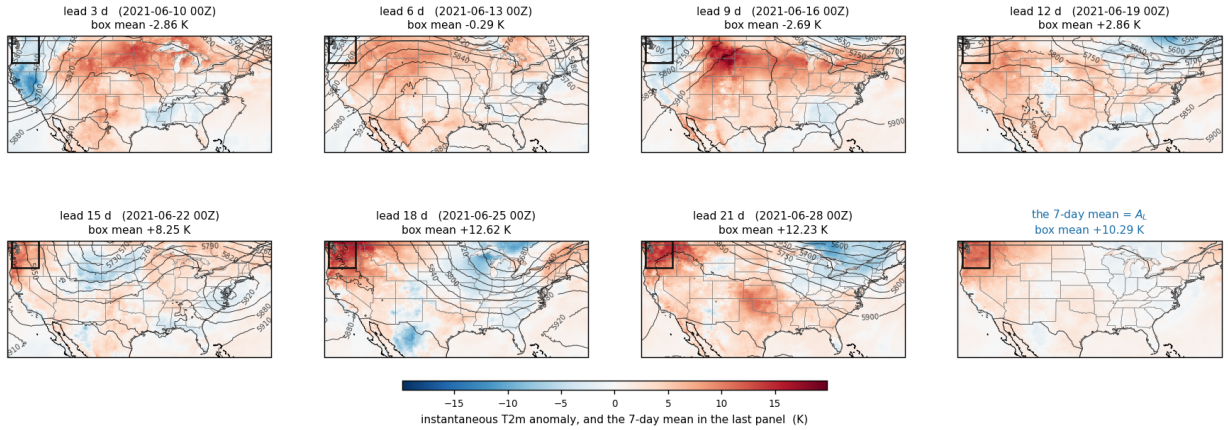


Figure 2: The single hottest surviving trajectory of the 2021 heat-dome run, followed from the 3-day checkpoint to the event, with the instantaneous temperature anomaly shaded and the upper-atmosphere pressure pattern contoured. At the 9-day checkpoint this trajectory had the fourth-coolest surface temperature of the 64 alive and would have ranked near the bottom on the conventional score. The scoring model ranked it tenth and gave it the maximum of eleven clones, because the ridge that later drives the heat dome was already building offshore while the surface beneath it had not yet warmed.

level rather than beyond it. Past that boundary the limit is population size, which scales linearly in compute. Nine cases do not establish a hit rate, and the percentile above should be read as a working expectation rather than as a measured threshold.

Figure 1 shows the 2021 heat dome as fields rather than as a single number. The AI+RES population reproduces the dome in the observed location and at the observed intensity, and the direct ensemble average from the same starting point contains no dome. Figure 2 follows the hottest surviving trajectory of that run through the resampling steps and shows what the scoring model was responding to.

4 Cost and speed

The alternative route to trajectories that reach the event is to run a large plain ensemble until some of its members land there by chance. Table 3 prices that route. It takes the probability AI+RES estimated for each event, asks how many plain forecast runs would be needed to expect the same number of members on the event, and multiplies by the measured cost of one 21-day run, about 0.4 H100-hours or \$1.00.

The saving in Table 3 is not the primary result. At the \$156 budget of Table 2, direct sampling returns no members at all on these events, which is the measured 0 of 7 in Table 1. The cost multiple grows with rarity, from 5x on the least rare event priced here to 23x on the rarest, because direct-sampling cost scales as the reciprocal of the event

Table 2: One AI+RES forecast, measured, on one node of eight NVIDIA H100 GPUs. Dollar figures use \$2.50 per GPU-hour, or \$20 per hour for the node, a mid-market on-demand cloud rate.

hardware	1 node, 8x NVIDIA H100 (80 GB)
settings	64 trajectories, 5 resampling steps every 3 days, 6 scoring runs per trajectory
forecast horizon	21 days, scored over the final 7-day window
wall-clock, one forecast	7.2 to 8.7 hours
compute, one forecast	61 to 64 H100-hours
cost, one forecast	\$153 to \$160
throughput	about 3 forecasts per node per day

Table 3: *Estimated, not measured.* Direct-sampling cost to obtain the same tail sample AI+RES delivered, against the \$156 AI+RES run of Table 2. Derived from the AI+RES probability estimate for each event and a measured per-run cost. No ensemble of this size was run, and the probability estimates themselves carry about a factor of two at 64 trajectories, so the cost multiples carry the same factor.

event	members AI+RES put on target	event probability	plain runs needed (est.)	direct-sampling cost (est.)	cost multiple
Pacific NW heat dome, Jun 2021	42 of 64	1-in-19	780	\$780	5x
California heat wave, Sep 2022	25 of 64	1-in-53	1,320	\$1,320	8x
S. Texas heat dome, Jun 2023	6 of 64	1-in-370	2,220	\$2,220	14x
Winter Storm Elliott, Dec 2022	20 of 64	1-in-127	2,530	\$2,530	16x
Winter Storm Uri, Feb 2021	32 of 64	1-in-111	3,560	\$3,560	23x

probability while AI+RES cost is flat in it. The 2020 Southwest heat wave is absent from Table 3 because neither method reached it. Direct sampling would need an estimated 3,300 runs, about \$3,300, for a single member, and AI+RES at this budget did not reach it either.

5 What the result supports

The result supports a specific claim. On nine out-of-sample cases, concentrating a fixed compute budget on the trajectories heading toward an extreme produces forecast members on events that direct sampling misses entirely at the same or a larger budget, at a measured \$153 to \$160 and 7.2 to 8.7 hours per forecast.

It does not support treating these numbers as a calibrated probability service. Three limits qualify the result.

1. **Nine events do not establish a hit rate.** The 6-of-7 figure of Table 1 carries a wide interval. A 42-case hindcast campaign designed to produce a reliability curve is built and staged, and has not been run.
2. **The probabilities carry about a factor of two.** They are internally consistent, in that where direct sampling can resolve a probability our estimate agrees with it within sampling error, but the absolute level at 64 trajectories is noisy. The method also returns no usable probability for an event milder than the population it has already concentrated, which is a property of the algorithm rather than a failure of a particular run.
3. **The sampled distribution is the forecast model’s, not the atmosphere’s.** The supported claim is that the method reaches the extreme tail of a current forecast model efficiently. Whether that model’s tail is unbiased against the real climate is a separate question, and this work does not settle it.

Three lines of work follow. The 42-case calibration campaign, at an estimated \$1,700 of GPU time for its first 12-case tier, would replace the qualitative bound of Section 3 with a reliability curve. A scoring function for tropical cyclones is needed because the weekly regional average used here averages away a one-day landfall. Lighter model pairs would reduce the per-forecast cost directly, since the current pair requires an 80 GB GPU per trajectory.

A Appendix. Every trajectory, for four events

These figures show every forecast trajectory of four AI+RES runs, from the start of the forecast to the event. They support Table 1 and are not counted in the four pages of the brief. Figure A1 describes the panels, and Figures A2 to A4 use the same layout.

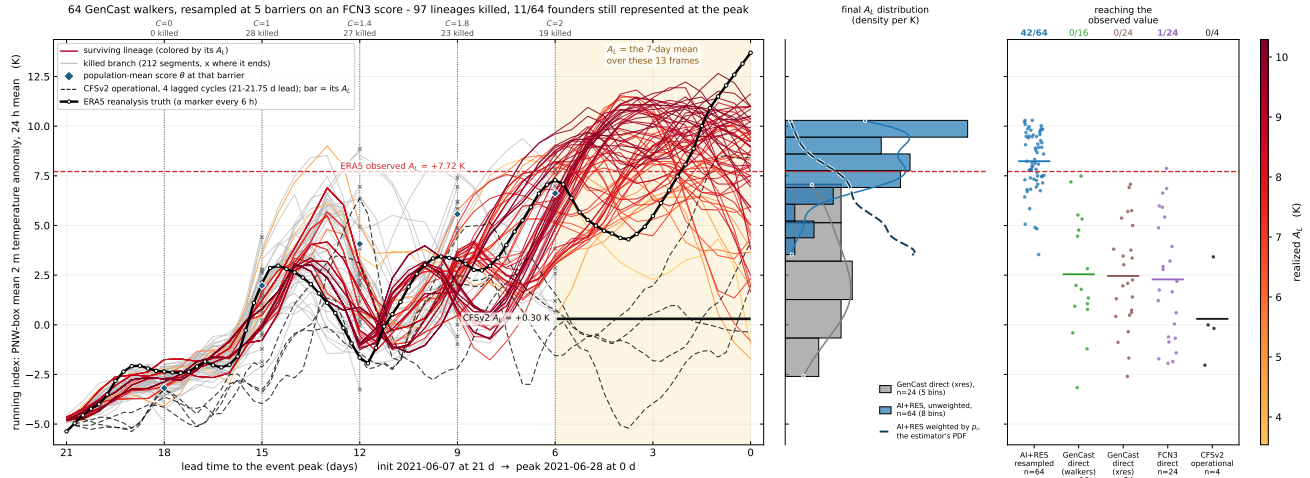


Figure A1: The 2021 Pacific Northwest heat dome, 64 trajectories, forecast from 21 days out. **Left.** Each line is one trajectory, plotted as its running regional temperature anomaly against time. Grey lines were discarded at a resampling step. Coloured lines survived to the event and are shaded by the value they reached. The dotted verticals are the five resampling steps, labelled by how hard the population was pushed and by how many trajectories were discarded. The blue diamonds are the population-average score. The shaded band on the right is the week the forecast is scored over. The black line with markers is the observed weather. The black dashed curves are the four CFSv2 operational cycles at the same lead, with their average drawn as a bar across the scored week. **Middle.** The distribution of final values for the 24-member direct GenCast ensemble in grey, for the AI+RES population in blue, and for the weighted estimate as a dashed line. **Right.** Where each method's members landed, with the number that reached the observed value printed above each column. Observed +7.72 K, reached by 42 of 64 trajectories, most extreme +10.29 K. The direct ensemble reaches it with none of 24 and stops at +7.08 K.

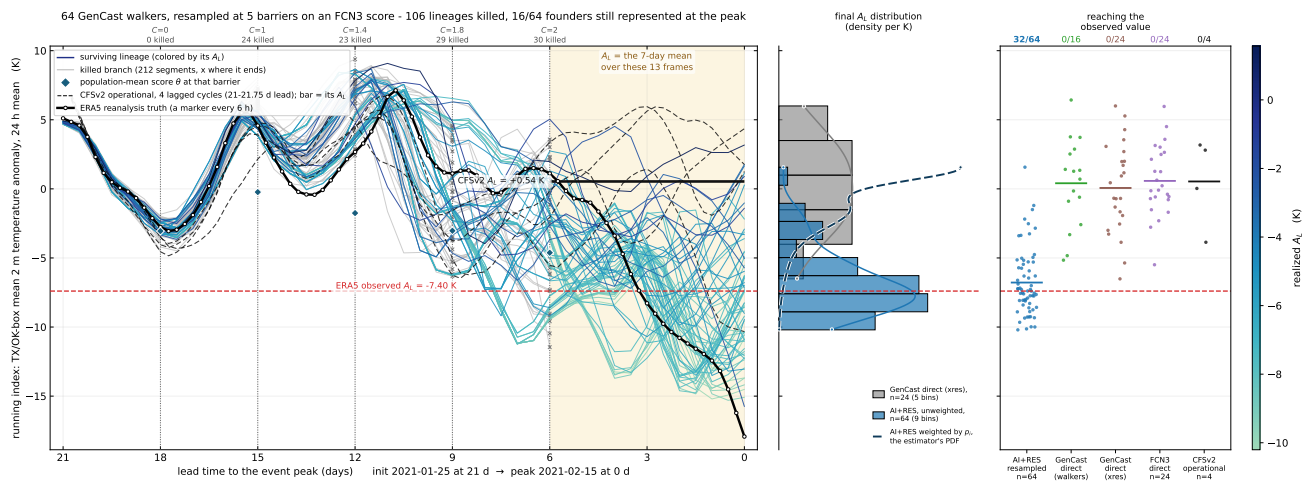


Figure A2: Winter Storm Uri, February 2021, over the Texas and Oklahoma region. The sign of the score is reversed for a cold event, so the method clones the coldest trajectories rather than the warmest. The schedule is otherwise the one used for the heat waves and was not retuned. Observed -7.40 K, reached by 32 of 64 trajectories, coldest -10.20 K. None of 24 direct GenCast members, 16 free-running trajectories or 24 direct FourCastNet 3 members reaches it.

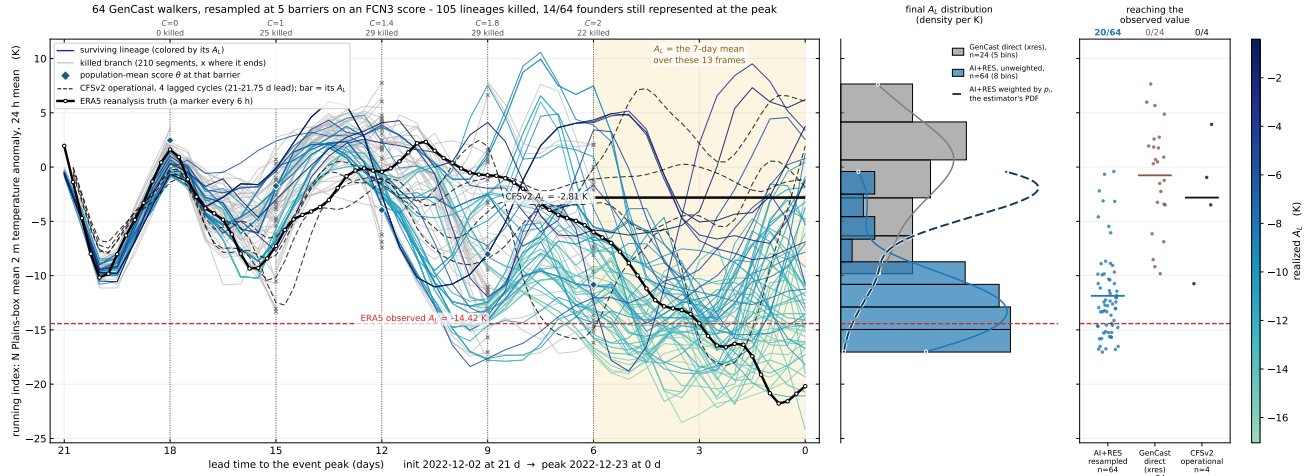


Figure A3: Winter Storm Elliott, December 2022, over the northern Plains region. This is the deepest cold target in the set relative to the forecast model's own spread. Observed -14.42 K, reached by 20 of 64 trajectories, coldest -17.04 K. The coldest direct member stops 4.6 K short at -9.83 K.

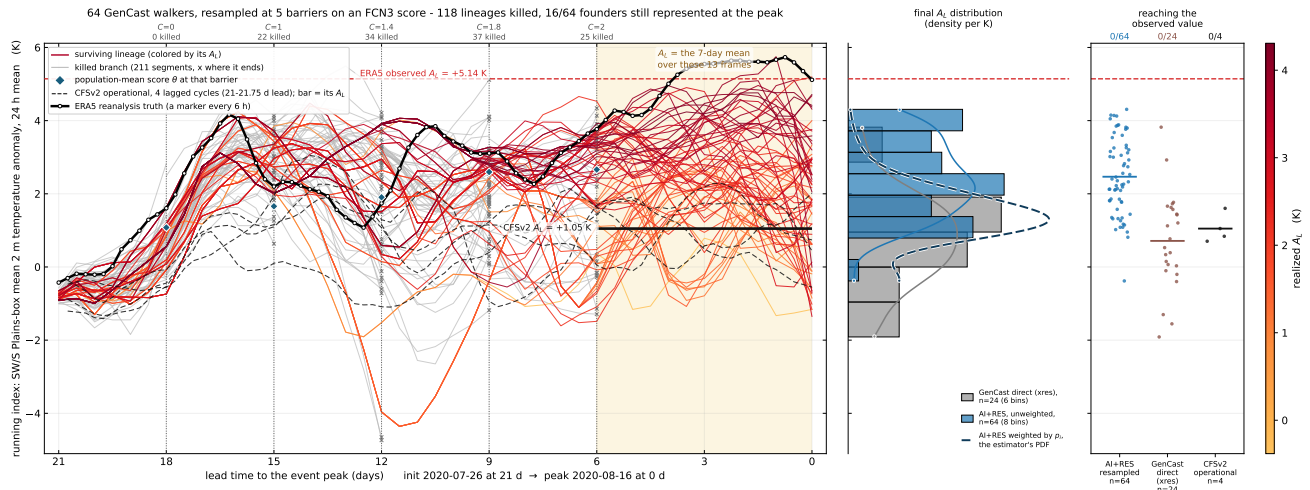


Figure A4: The August 2020 Southwest heat wave, the one target in Table 1 the method does not reach. Resampling moves the population about 1.75 K toward the observed value and ends 4.4 K short of it at $+4.31$ K. Where direct sampling and the weighted estimate both have members, they agree, so the miss is a limit on how far 64 trajectories can be pushed and not a failure of the weighting.